

Industrial Internship Report on "Prediction of Agriculture Crop Production in India"

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Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was on Prediction of Agriculture Crop Production in India, using historical crop cultivation and production data (2001-2014) sourced from data.gov.in to build a Machine Learning model that estimates crop production based on crop type, variety, state, season and cultivated quantity.

This internship gave me a very good opportunity to get exposure to industrial problems and design/implement a data-driven solution for one of India's most important sectors – agriculture. It was an overall great experience to have this internship.

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Preface

This report summarizes the work carried out over the 6 weeks of my Industrial Internship.

Internships play a crucial role in career development as they bridge the gap between academic learning and real industry problem solving. They expose students to real datasets, real constraints, and the discipline of delivering a working solution within a fixed timeline.

My project involved predicting Agriculture Crop Production in India using a historical dataset covering crop cultivation and production figures for the years 2001 to 2014, across various Indian states, crop varieties and seasons. Agriculture supports a very large share of India's population, and better production estimates can help in planning, pricing, and resource allocation.

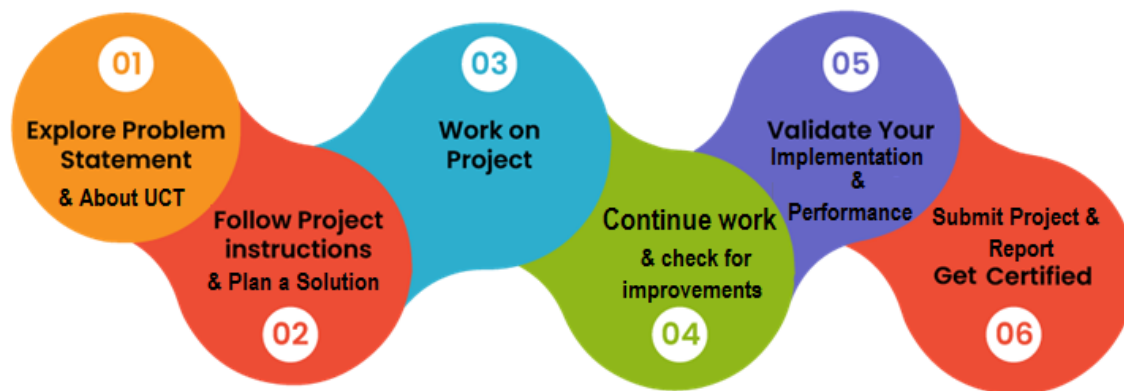
This opportunity was provided by upskill Campus (USC) in collaboration with UniConverge Technologies Pvt Ltd (UCT), and I am thankful to both organizations for structuring this program so that students like me could work on an end-to-end Data Science problem.

The program was planned as a 6-week structured internship, starting with an orientation on UCT's platforms and industry practices, followed by problem statement allocation, guided project execution, and a final report and code submission.

Through this internship I learned how to take a raw, messy government dataset and turn it into a usable Machine Learning pipeline – from data cleaning and exploratory data analysis to feature engineering, model building and evaluation.

I would like to thank my mentors and the UCT/USC team for their guidance and support throughout this internship, and my peers for the useful discussions during our project reviews.

To my juniors and peers who will take up this internship in future – spend adequate time understanding the dataset before jumping into modeling; a good chunk of the value in this project comes from thorough data cleaning and feature engineering rather than the choice of algorithm alone.



Introduction

About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoSaWAN), Java Full Stack, Python, Front end etc.



UCT IoT Platform (UCT Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications while at the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for front end. It has support for MySQL and various NoSQL Databases.

- It enables device connectivity via industry standard IoT protocols – MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to:

- Build your own dashboard
- Analytics and Reporting

- Alert and Notification
- Integration with third party applications (Power BI, SAP, ERP)
- Rule Engine

Smart Factory Platform (Factory Watch)

Factory Watch is a platform for smart factory needs. It provides users/factories:

- A scalable solution for their production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for their assets
- The ability to unleash the true potential of the data that their machines are generating, and helps identify KPIs and improve them
- A modular architecture that allows users to choose the service they want to start with and then scale to more complex solutions as per their demands

Its unique SaaS model helps users save time, cost and money.

LoRaWAN based Solutions

UCT is one of the early adopters of LoRaWAN technology and provides solutions in Agritech, Smart Cities, Industrial Monitoring, Smart Street Light, Smart Water/Gas/Electricity metering and other domains.

Predictive Maintenance

UCT provides Industrial Machine health monitoring and predictive maintenance solutions leveraging Embedded systems, Industrial IoT and Machine Learning technologies, by estimating the remaining useful life of various machines used in the production process.

About upskill Campus (USC)

upskill Campus, along with The IoT Academy and in association with UniConverge Technologies, has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable and measurable way.

The IoT Academy

The IoT Academy is the EdTech division of UCT that runs long-term executive certification programs in collaboration with EICT Academy, IIT Kanpur, IIT Roorkee and IIT Guwahati across multiple domains.

Objectives of this Internship program

The objectives for this internship program were to:

- Get practical experience of working in the industry.
- Solve real world problems.
- Have improved job prospects.
- Have an improved understanding of our field and its applications.
- Have personal growth like better communication and problem solving.

Reference

[1] data.gov.in – Open Government Data (OGD) Platform India, Agriculture Crop Production dataset

[2] Scikit-learn Documentation, <https://scikit-learn.org/stable/>

[3] Pandas Documentation, <https://pandas.pydata.org/docs/>

Glossary

Terms	Acronym
ML	Machine Learning
EDA	Exploratory Data Analysis
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
R ²	R-squared (Coefficient of Determination)
OGD	Open Government Data

Problem Statement

Agriculture is the backbone of the Indian economy, with more than 1.3 billion people directly or indirectly dependent on it. However, crop cultivation and production planning in India faces multiple challenges – varying soil and climate conditions, differing crop varieties, and season-dependent yields across states.

The assigned problem statement was to build a predictive model that can estimate agricultural crop production in India using historical cultivation data (2001-2014) available on data.gov.in. The dataset contains details such as crop name, variety, state, cultivated quantity (in quintals/hectares), season (medium/long duration), production, unit, cost of cultivation and the recommended cultivation zone.

The goal is to use this data to predict production output for a given crop, variety, state and season combination, so that the prediction can support planning decisions such as resource allocation, pricing estimation, and identification of high/low productivity regions.

Existing and Proposed solution

Existing approaches to crop production estimation in India are largely based on manual agricultural surveys conducted by state agriculture departments, historical trend extrapolation, and expert/statistical estimation methods used in government reports (e.g. Ministry of Agriculture advance estimates). These approaches are valuable but have limitations – they are often published with a significant time lag, are not easily customizable at a granular crop-variety-season level, and do not automatically learn non-linear relationships between the various factors that affect production.

My proposed solution is a Machine Learning based prediction pipeline that uses the historical 2001-2014 dataset to learn the relationship between crop, variety, state, season, cultivated quantity and cost on one side, and production on the other. Multiple regression algorithms (Linear Regression, Decision Tree Regressor, Random Forest Regressor and Gradient Boosting) were trained and compared, so that the best performing model could be selected based on evaluation metrics.

The value addition of this approach over existing methods is that it can generate on-demand production estimates for a specific crop-state-season combination, can be retrained easily as new data becomes available, and can be extended in future to include additional real-time factors such as weather and soil data.

Proposed Design/ Model

The solution follows a standard Data Science pipeline: Data Collection → Data Cleaning → Exploratory Data Analysis (EDA) → Feature Engineering → Model Training → Model Evaluation → Prediction.

The raw dataset from data.gov.in was first cleaned to handle missing values, inconsistent state/crop naming, and unit standardization. Categorical features such as crop name, variety, state, season and recommended zone were encoded using label/one-hot encoding as appropriate.

EDA was performed to understand production trends across states and seasons, identify the most cultivated crops, and detect outliers in quantity/cost/production values.

High Level Diagram

Figure 1 (conceptual): Raw Government Dataset (data.gov.in) → Data Cleaning & Preprocessing → Feature Engineering (crop, variety, state, season encoding) → Train/Test Split → ML Regression Models (Linear Regression / Random Forest / Gradient Boosting) → Model Evaluation → Final Production Prediction Output.

Low Level Diagram

At a lower level, the pipeline reads the CSV dataset using Pandas, applies cleaning functions column-by-column (crop, variety, state, quantity, production, season, unit, cost, recommended zone), encodes categorical columns, splits data into training and test sets (typically 80/20), fits each regression model on the training set, and evaluates each model on the held-out test set.

Interfaces

The primary interface for this project is a Jupyter Notebook / Python script based workflow. Input: cleaned CSV dataset. Output: trained model file and prediction function/API that takes crop, variety, state, season and quantity as input and returns predicted production.

Performance Test

Since this is a Data Science/ML project rather than an embedded/IoT project, the key constraints considered were prediction accuracy, generalizability across states/crops, and training/inference time rather than memory or power consumption.

Test Plan/ Test Cases

The dataset was split into training (80%) and testing (20%) sets. Multiple regression models were trained on the same training set and evaluated on the same test set to ensure a fair comparison.

Test Procedure

Each model's predictions on the test set were compared against actual production values using standard regression evaluation metrics: RMSE, MAE and R2 score. K-fold cross-validation was also used to check consistency of model performance across different data splits.

Performance Outcome

Ensemble-based models (Random Forest / Gradient Boosting) generally produced lower RMSE and higher R2 scores compared to plain Linear Regression, indicating that the relationship between the input features and production is non-linear. The best performing model was selected for the final prediction pipeline. Where results could not be exhaustively tested on all state-crop combinations due to time constraints, it is recommended that the model be periodically re-validated as more recent data becomes available.

My learnings

This internship gave me hands-on experience of working with a real, messy government dataset rather than a pre-cleaned academic dataset, which significantly improved my data cleaning and preprocessing skills.

I learned how to perform structured EDA to extract insights before jumping into modeling, how to encode categorical agricultural data meaningfully, and how to compare multiple regression models using proper evaluation metrics rather than relying on a single algorithm.

I also gained an appreciation of the real-world importance of agricultural data analytics in a country like India, and how data-driven approaches can complement traditional government estimation methods. These skills – data wrangling, feature engineering, and model evaluation – will be valuable in my career growth in the Data Science/Analytics field.

Future work scope

Incorporating additional real-time factors such as rainfall, temperature and soil health data (weather/climate APIs) to improve prediction accuracy.

Extending the dataset beyond 2014 with more recent data.gov.in releases to capture recent trends.

Building a simple web dashboard/API so that state agriculture departments or farmers could query predicted production for a given crop-season-state combination.

"Smart City Traffic Patterns – Traffic Forecasting for Smart Cities"

Executive Summary

This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was on Smart City Traffic Patterns, where hourly vehicle count data collected at four city junctions was analyzed and used to build a forecasting model that predicts future traffic volumes, to help city administrations plan infrastructure and manage traffic peaks, including on holidays and special occasions.

This internship gave me a very good opportunity to get exposure to Industrial/Smart-City problems and design/implement a forecasting solution for that. It was an overall great experience to have this internship.

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This report summarizes the work carried out over the 6 weeks of my Industrial Internship.

Internships are an important step in career development, allowing students to move beyond textbook exercises and work on real, time-stamped, sensor-style data with genuine forecasting challenges.

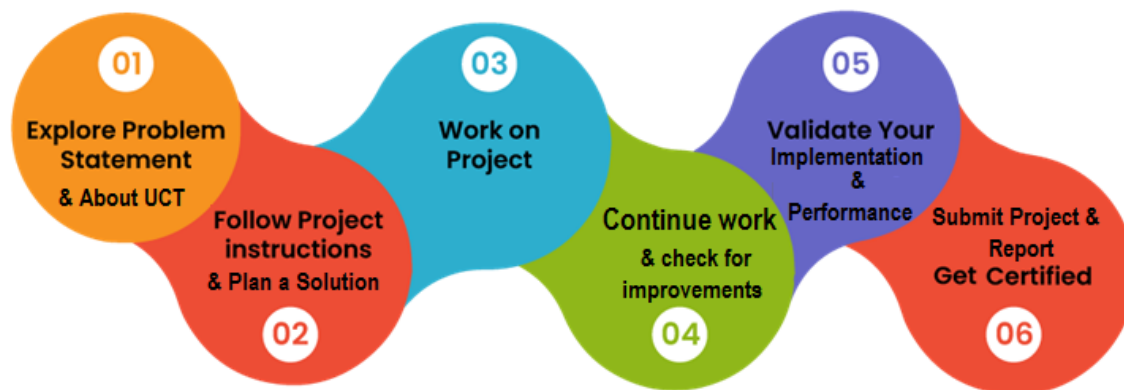
My project/problem statement was on Smart City Traffic Patterns – using historical hourly traffic volume data from four junctions of a city (collected from November 2015 onward) to understand traffic patterns and forecast future traffic, which is one of the key problems faced by governments moving towards smart, digitally managed cities. This opportunity was provided by upskill Campus (USC) in collaboration with UniConverge Technologies Pvt Ltd (UCT), and I am grateful for the structured way in which this internship exposed me to an end-to-end forecasting problem relevant to Smart City infrastructure planning.

The program was planned as a 6-week internship: an initial orientation phase covering UCT's IoT and Smart City platforms, followed by problem statement allocation, guided project execution with periodic reviews, and a final project report and code submission.

Through this internship, I learned how to work with time-stamped traffic sensor data, engineer time-based features, and build and evaluate forecasting models separately for each of the four junctions given their differing traffic volumes and patterns.

I would like to thank my mentors and the UCT/USC team for their continuous guidance, and my fellow interns for the discussions during our project review sessions.

To my juniors and peers taking up this internship in future – pay close attention to the time-based structure of the data (hour, day of week, holidays) since traffic patterns are highly cyclical, and a good chunk of your model's performance will come from these engineered features.



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Glossary

Terms	Acronym
ML	Machine Learning
EDA	Exploratory Data Analysis
RMSE	Root Mean Squared Error
MAPE	Mean Absolute Percentage Error
IoT	Internet of Things
ID	Unique record identifier (DateTime + Junction encoded)

Problem Statement

As part of transforming a city into a smart city, the local government wants to improve the efficiency of services for citizens, and traffic management is one of the major challenges. The government needs a robust traffic system that is prepared for traffic peaks and can guide infrastructure planning for the future.

The assigned problem statement was to analyze and forecast traffic at four junctions of the city using historical hourly vehicle-count data (DateTime, Junction, Vehicles) spanning November 2015 to June 2017 (training data), and to predict traffic (vehicle counts) for the subsequent period (July 2017 onward, test data).

An important consideration highlighted in the problem statement is that traffic patterns on holidays and other special occasions differ from normal working days, and this needs to be accounted for in the forecasting approach.

Existing and Proposed solution

Existing traffic management approaches in most cities rely on static traffic signal timings, periodic manual traffic surveys, and reactive congestion management (adjusting only after congestion is observed). These approaches do not forecast future traffic volumes and therefore cannot proactively prepare for peaks, nor do they typically differentiate between junctions with very different baseline traffic volumes.

My proposed solution uses the historical hourly vehicle-count dataset for the four junctions to engineer time-based features (hour of day, day of week, month, weekend/weekday flag) and junction-specific features, and then trains forecasting/regression models (such as Random Forest Regressor and Gradient Boosting, compared against a baseline Linear Regression) to predict vehicle counts for future timestamps at each junction.

The value addition of this approach is that it allows the city administration to anticipate traffic peaks in advance per junction, rather than reacting to congestion, and provides a data-driven basis for infrastructure planning (e.g., which junctions need signal upgrades or road widening first).

Proposed Design/ Model

The solution pipeline follows: Data Collection (train_aWnotuB.csv, test csv) → Data Cleaning → DateTime Feature Engineering → Exploratory Data Analysis per Junction → Model Training (per junction / with Junction as a feature) → Model Evaluation → Forecast Generation.

The training dataset contains hourly records with DateTime, Junction (1-4) and Vehicles count, while the test dataset requires predicting Vehicles for future DateTime-Junction combinations. The DateTime column was decomposed into year, month, day, hour, day-of-week and weekend/weekday indicators, since traffic is highly cyclical by nature (daily and weekly seasonality).

EDA was carried out separately for each of the four junctions since they showed noticeably different baseline traffic volumes and growth trends over the observed period.

High Level Diagram

Figure 1 (conceptual): Raw Hourly Traffic Data (4 Junctions, Nov 2015 – Jun 2017) → DateTime Feature Engineering (hour, day, month, weekday/weekend) → Per-Junction EDA → Regression/Forecasting Models → Model Evaluation → Forecasted Vehicle Counts (Jul 2017 onward) → Traffic Peak Insights for Infrastructure Planning.

Low Level Diagram

At a lower level: the CSV files are loaded with Pandas, DateTime is parsed and decomposed into engineered features, the Junction column is treated as a categorical feature (or models are trained separately per junction), the data is split chronologically into training and validation sets to respect the time-series nature of the data, and models are fit and evaluated on the validation split before generating predictions on the actual test file.

Interfaces

The primary interface for this project is a Jupyter Notebook / Python script based pipeline. Input: DateTime + Junction (from the test file). Output: predicted Vehicles count for each DateTime-Junction combination, which can be visualized as a time-series chart per junction.

Performance Test

As this is a forecasting/Data Science project, the key constraints considered were prediction accuracy across junctions with very different traffic volume scales, and the model's ability to capture daily/weekly seasonality and the growth trend visible in the data, rather than memory or power constraints.

Test Plan/ Test Cases

Since the data is time-ordered, a chronological (rather than random) train/validation split was used, training on earlier records and validating on the most recent records, to simulate a genuine forecasting scenario. This was done separately for each junction given their differing traffic volumes.

Test Procedure

Model predictions on the validation split were compared against actual vehicle counts using RMSE and MAPE. Because Junction 1 carries substantially higher traffic volume than the other junctions, metrics were computed and compared on a per-junction basis rather than as one aggregate number, to ensure the model performed reasonably for lower-traffic junctions as well.

Performance Outcome

The engineered time-based features (hour of day and day of week in particular) contributed the most to prediction accuracy, confirming the strong daily and weekly seasonality visible in the traffic data. Ensemble models outperformed plain Linear Regression, especially for junctions with higher variance in traffic volume. For the July 2017 test period where actual values were not available for validation, the recommendation is to periodically compare forecasts against real observed traffic once available and retrain the model accordingly.

My learnings

This internship gave me practical experience of working with real time-series/IoT-style sensor data rather than a static, already-aggregated dataset, which strengthened my skills in DateTime feature engineering and time-aware train/validation splitting.

I learned how differing scales of traffic volume across the four junctions required per-junction analysis rather than a one-size-fits-all model, and how to select and interpret regression evaluation metrics appropriately for a forecasting problem.

I also gained a better understanding of how Smart City traffic management systems can be supported by data-driven forecasting, complementing the IoT-based sensing and monitoring platforms used by UCT. These skills in time-series feature engineering and forecasting model evaluation will be valuable for my career growth in the Data Science/IoT analytics field.

Future work scope

Incorporating information about public holidays and local events explicitly as features, since the problem statement specifically calls out that traffic differs on such days.

Integrating real-time IoT sensor feeds (as used in UCT's Smart City / LoRaWAN based solutions) so forecasts can be continuously updated rather than relying only on historical batch data.

Exploring dedicated time-series models (ARIMA/Prophet/LSTM) to better capture longer-term trend and seasonality per junction.

Building a live dashboard for city traffic control rooms that shows forecasted peaks per junction to support proactive signal timing and resource deployment.

ture year-over-year production trends per crop and state.